

Automated Leukaemia Prediction and Classification using Deep Learning Techniques

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Abstract: Leukemia is typically diagnosed based on an abnormal blood count, frequently an elevated White Blood Cell (WBC) count. The diagnosis is established through bone marrow, replaced by neoplastic cells. Acute Lymphoblastic Leukemia (ALL) is a type of leukaemia that affects the blood and bone marrow. Leukaemia primarily affects children and adults around the world. Early leukaemia detection is critical for appropriately treating patients, especially children. This research aims to present a diagnostic method that uses computational intelligence and image processing algorithms to identify blast cells from ALL images. The medical image is prepared initially using the preprocessing and segmentation technique for efficient classification. In this research, the type is accomplished using Bidirectional Associative Memory Neural Networks (BAMNN), where the accuracy is 96.87%, the highest classification rate and outperforms the existing technique.

Keywords: Leukaemia, DL, neural network, classification, accuracy, activation function, neurons, bone marrow, and associative memory.

I. INTRODUCTION

Many blood-related disorders can be diagnosed in part by visually examining blood smears. A precise diagnosis and prompt treatment can be achieved through the initial smear results and early leukaemia identification [1]. With minor errors and quick turnaround time, blood smear image analysis using Machine Learning (ML) algorithms may diagnose early-onset leukaemia and identify subtypes. Future research may use cutting-edge ML, particularly Deep Learning (DL), in CAD systems, whole-slide imaging (WSI), and even mobile applications and software in haematology labs to aid pathologists and oncologists in correctly identifying leukaemia [2-3]. An increase in white blood cells in the bone marrow and peripheral circulation frequently characterizes malignant illnesses of the hematopoietic stem cell compartment known as leukaemias [4]. Depending on the type, leukaemia's course might last anywhere from a few days or weeks to several years. A malignant tumour that affects the bone marrow and blood is

called leukaemia. It results from the body's abnormal synthesis of white blood cells. Millions of cells are combined to produce the liquid substance known as blood [5]. White Blood Cells (WBCs), platelets, Red Blood Cells (RBCs), and plasma are included among its numerous blood cell constituents.

WBCs comprise just 1% of the blood's anti-infection capacity in the human body [6-8]. The elements of these WBCs that are most impacted by cancer. 40% to 45% of RBCs in blood volumes transport oxygen to the body. Smaller cells called platelets aid in the regulation of bleeding. Plasma is the term for the blood's liquid component [9-10]. It aids in delivering nutrients and water to the tissues. "bone marrow" refers to the entire collection of blood components produced in the blood factory.

In the case of leukaemia, the bone marrow makes many atypical, immature white blood cells [11-12]. Leukaemia is diagnosed in three ways, namely, physical tests, blood tests, and bone marrow tests. In the physical examination, the haematologist reviews the complete medical records and health history and checks for the presence of pale skin, enlarged spleen, and bruising in the human body [13]. Total Blood Count (CBC) was measured in a blood test to find the count of WBCs, RBCs, and platelets. The peripheral blood smear test is conducted to identify the blood cells' shape, which helps diagnose leukaemia. In a bone marrow test, the bone marrow part is collected from the hipbone and used to determine the characteristics of leukaemia cells. The types of treatment for leukaemia depend on many factors, such as age, type of leukaemia, and extent or spread of leukaemia cells in the body [10-14-18]. Leukaemia is typically diagnosed based on an abnormal blood count, frequently an elevated White blood cell count [19-20]. It enables a precise diagnosis and valid predictive data, enabling therapy to be customized to the patient's disease. For an accurate diagnosis of all, lymphoblast in stained peripheral blood and bone marrow samples must be successfully identified and subtyped [21]. Additionally, it is crucial to classify blasts by their subtype because this

information is essential for determining the prognosis and the best course of chemotherapy.

The French, American, and British (FAB) joint classification system and the World Health Organization's (WHO) recommended nomenclature are the foundation for leukaemia subcategorization standards. As it considers morphological, immunological, phenotypic, cytogenetic, and molecular aspects in evaluating leukemic blasts and is more significant to therapeutic or prognostic implications, the WHO classification is more widely accepted than the FAB system. However, the WHO Classification is more complex than the FAB system due to the additional evaluation of blasts using a flow cytometer and genetic analysis. Additionally, the higher cost deters the use of flow cytometers for routine ALL screening in many healthcare facilities throughout many developing nations. These elements make the conventional practice for screening and sub-typing of ALL, which is a microscopic inspection of blood samples, necessary [22]. The Leukaemia classification uses ML and DL classification techniques. ML techniques play a prominent role in identifying the occurrence of leukaemia. The hybridized ML techniques are efficient in handling the leukaemia data. Linear Regression-based Support Vector Machine (LDSVM) plays a prominent role in impact and disease analysis. The segmentation is achieved for an efficient clustering technique where k-means clustering is efficient [23]. DL technique Convolutional Neural Networks (CNN) and Deep Neural Networks (DNN) are significant in the classification of Leukaemia data [24].

A. Motivation

Modern imaging techniques like segmentation, feature extraction, and classification aid accurate diagnosis and treatment planning. The inspiration for the current research comes from the potential of computational intelligence and its application in the medical field. This research aims to introduce innovative approaches to precisely India and other Low and Middle-Income nations that account for most leukaemia mortality, where most patients are diagnosed at later stages of their disease.

Diagnosing ALL early is crucial, facilitating appropriate treatment to enhance the effective recovery of patients, especially children. Moreover, due to advancements in treatment modalities, proper ALL staging helps physicians plan suitable treatment. Most research articles concentrate on various algorithms and techniques for segmenting white blood cells and classifying normal lymphocytes from abnormal lymphoblasts. To aid the clinical expert in the proper diagnosis and treatment of ALL, a system is developed that not only confirms the presence of ALL but also specifies the stage of the disease, according to the French American British (FAB) System, which makes it easy for Physicians to give appropriate treatment and increase the chance of recovery.

B. Research Gap

The development of more sophisticated medical imaging techniques has made it exceedingly challenging to assess the images manually. A sophisticated and effective CAD system is

required to analyze the images, interpret them, and identify the disorders. Malignant malignancy of the blood and bone marrow is known as leukaemia. Leukaemia may be divided into main classifications: acute and chronic. Acute Lymphoblastic Leukaemia (ALL) and Acute Myeloid Leukaemia (AML) are the two subtypes of acute leukaemia. Most children with ALL paediatric cancer are between ages 5 and 14. Suppose leukaemia is anticipated and detected in its early stages. In that case, the patient can receive the proper therapy, which can cure the disease, hasten recovery, and save the lives of many leukaemia patients. Cells in leukaemia are intricate. Leukaemia disease diagnosis by medical pictures is a challenging undertaking. Blood cells are complex, arising from the diversity in cell size, shape, appearance, and blood composition. Using the size and form of the leukaemia nucleus picture, non-blast and blast cells are distinguished. Therefore, nucleus segmentation in the images is crucial.

C. Research Objective

This research aims to provide a diagnostic method that uses computational intelligence and image processing algorithms to identify ALL blast cells. An experimental study utilizes relevant image processing and computational intelligence approaches to choose the best method for delineating ALL compartments. To forecast leukaemia, the following goals have been set forward,

- The medical image is resized to maintain the quality of all images equally, and the medical artefacts are removed with the support of Gaussian blur.
- Further, the image is segmented using the Fuzzy C-Means Clustering method, where the significant regions are highlighted.
- Finally, leukaemia is classified using Bidirectional Associative Memory Neural Networks (BAMNN), where regular and abnormal images are organised.

The leukaemia classification article is organized as follows: the overview, discussion about state-of-art techniques, motivation, and gap analysis are given in Section 1, the proposed Bidirectional Associative Memory Neural Networks (BAMNN) based Leukemia Classification is detailed in Section 2, the overall result and its discussion with depiction is given in Section 3, the article is concluded with the future recommendation in Section 4.

II. PROPOSED METHODOLOGY- BIDIRECTIONAL ASSOCIATIVE MEMORY NEURAL NETWORKS (BAMNN) BASED LEUKEMIA CLASSIFICATION

This section explains the preprocessing, segmentation, and classification technique where Gaussian blur is utilized in the preprocessing for removing the noise. The prominent regions and the edge portions are highlighted using Fuzzy C-Means Clustering (FCM), and the classification is by using Bidirectional Associative Memory Neural Networks (BAMNN) [25].

A. Preprocessing

Pre-processing leukaemia images involves a series of steps to enhance the quality and extract relevant features from the images before feeding them into a ML model for analysis. Resize the images to a consistent resolution to ensure uniformity in the test dataset. This step is essential to reduce computation requirements and improve model performance. Normalize the pixel values of the images to a standard scale (e.g., [0, 1] or [-1, 1]). It helps in stabilizing training and improves convergence during the model training process. Gaussian blurring is applied to remove unwanted noise from the images.

Gaussian blur is a commonly used image filtering technique in image processing. It's a low-pass filter that helps reduce high-frequency noise in an image while preserving its overall structure. The name "Gaussian" comes from the mathematical function used to calculate the weights of neighbouring pixels during the blurring process. The mathematical model for applying Gaussian blur to an image involves convolving the image with a Gaussian kernel. The convolution operation is a mathematical operation that combines two functions to produce a third. Image processing involves sliding a kernel (filter) over the image and computing the weighted sum of pixel values. Here's the mathematical representation of applying Gaussian blur to an image using convolution:

Given an input image $I(x, y)$ and a Gaussian kernel $G(x, y)$, the Gaussian-blurred image $B(x, y)$ is obtained through convolution as Eq. (1):

$$B(x, y) = I(x, y) \times G(x, y) \quad (1)$$

Where:

$B(x, y)$ is the blurred image's pixel value at coordinates (x, y) .

$I(x, y)$ is the original input image's pixel value at coordinates (x, y) .

$G(x, y)$ is the Gaussian kernel centred at (x, y) .

The Gaussian kernel $G(x, y)$ is defined as, Eq. (2)

$$G(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2+y^2}{2\sigma^2}\right) \quad (2)$$

where ' σ ' is the standard deviation of the Gaussian distribution, controlling the amount of blur. A larger ' σ ' leads to a wider distribution and more blur.

The convolution operation involves sliding the Gaussian kernel over each image pixel and computing the weighted sum of pixel values the kernel covers. The resulting value becomes the new pixel value in the blurred image. The continuous Gaussian function is discretized to create a finite Gaussian kernel, which is then used for convolution. The size of the kernel (kernel width and height) determines the extent of blurring, with larger kernels resulting in stronger blur.

B. Segmentation

Segmenting leukaemia images using the Fuzzy C-Means (FCM) clustering algorithm involves partitioning the image into different regions or clusters based on the intensity values of the pixels. FCM is a soft clustering technique that assigns each pixel a membership value for each group, indicating the degree of membership to that cluster. The process of FCM is given in Algorithm 1.

Algorithm 1: Pseudo-code for Fuzzy C-Means Clustering

Input: Microscope ALL Images of Peripheral Blood

Output: Segmented ALL images.

Procedure

Step 1. Initialize ALL images

Step 2. Randomly select ' c ' cluster centres.

Step 3. Calculate the fuzzy membership. μ_{ab} using $\mu_{ab} = 1 / \sum_{m=1}^c k_{ab} / k_{am}^{(\frac{2}{n}-1)}$

Step 4. Compute initial cluster Centroids. $c_1, c_2, \dots, c_k$

Step 5. Compute the fuzzy centres. f_c using $f_c = (\sum_{m=1}^c (\mu_{ab})^n R_e) / (\sum_{m=1}^c (\mu_{ab})^n), \forall_b = 1, 2, \dots, n$

Step 6. Update Partition Matrix $(\mu_{ab})_{n \times c}$

Step 7. Repeat steps 3 and 5 until the minimum ' B ' value is achieved or $\|S^{(m+1)} - S^{(m)}\| < \beta$

where,

' m ' is the iteration step.

' β ' is the termination criterion between [0, 1].

' $S = (\mu_{ab})_{n \times c}$ ' is the fuzzy membership matrix.

' B ' is the objective function.

Step 8. Reshape the cluster pixels into an image

Step 9. End

C. Classification

Bidirectional associative memory is a form of RNNs proposed by Bart Kosko in the early 1980s to address various identified shortcomings of the auto-associative Hopfield NNs and ANNs, which learn associations of information through continuous training. The BAM stands for hetero-associative memory, which allows storing associated data of any type and structure without needing continual learning. The data associations are stored simultaneously before being recalled from the BAM's memory. Typically, the BAM surpasses the conventional ANNs, recalling associations with much superior performance.

In addition, as compared to existing ANNs, the BAM perfectly solves the known bitwise XOR problem, allowing the data to be stored in the binary (bipolar) form in its memory. Unlike the unidirectional Hopfield network, the BAM could recall the associations for data assigned to its inputs or outputs bidirectionally. It also permits the recovery of the correct associations for incomplete or corrupted data. When used as part of an AI-based decision-making process, BAM models can infer the solution to a particular data analysis problem based on numerous relationships of many interconnected data. Generally, the BAM models can be used for vast applications, such as (i) classification and clustering data, (ii) incomplete data augmentation, and (iii) recovering damaged or corrupted data.

When the variety of information acquired by ANNs is insufficient for processing data presented to an AI for analysis, the BAM models are beneficial. Bidirectional Associative

Memory Neural Networks (BAMNN) is a neural network model for associative memory tasks. It extends the concept of traditional Hopfield networks and Bidirectional Associative Memories (BAMs) to incorporate the capabilities of neural networks for both pattern recall and pattern completion. BAMNNs aim to improve the storage and retrieval of patterns in a bidirectional manner.

Associative Memory: Associative memory models are designed to store and retrieve patterns based on their associations. Given a partial or noisy input pattern, an associative memory should be able to recover the complete or most relevant associated pattern.

Bidirectional Associative Memory (BAM): A traditional BAM can associate pairs of patterns in forward and backward directions. Given a pattern in one layer (usually referred to as the "input" layer), the BAM can retrieve the associated pattern in the other layer (often called the "output" layer) and vice versa.

Neural Network Architecture: A BAMNN extends the BAM concept by incorporating the structure and capabilities of neural networks. It typically involves using layers of interconnected neurons, like feedforward neural networks. Each image is represented as a flattened vector of pixel values. The BAMNN architecture with an input layer (representing image pixels) and an output layer (representing class labels). Initialize the weights w for the input-to-output connections. The input layer, X , could use a linear activation function or a scaled sigmoid to map pixel values to the $[-1,1]$ $[-1,1]$ range. This work uses a binary step function or a sigmoid for output layer Y to obtain binary class predictions. During training, associate each input image with its corresponding class label using a modified Hebbian learning rule. Adjust the weights based on the product of input and output patterns. Weight update for w_{ij} (input-to-output), Eq. (3)

$$\Delta w_{ij} = \eta X_i Y_j \quad (3)$$

During training, present each image to the input layer and adjust the output layer based on the modified Hebbian learning rule. The weight 'w' learns to associate input patterns with class labels.

Pattern Recall and Completion: A key feature of BAMNNs is their ability to recall patterns (like traditional BAMs) and complete patterns. Given a partially corrupted or incomplete input pattern, the BAMNN can generate a coherent and comprehensive version of that pattern. Given an input image during the recall phase (classification), update the output layer based on the learned weights. For Y_j , Eq. (4)

$$Y_j = \text{sign}(\sum_{i=1}^N w_{ij} X_i) \quad (4)$$

Learning and Adaptation: BAMNNs often incorporate learning algorithms that allow the network to adapt and fine-tune its connections based on the patterns it is exposed to during training. Learning ensures that the network can better recall and complete patterns, even in noise or corruption.

Applications: BAMNNs have been explored for various applications, including pattern recognition, content addressable memory, and associative memory tasks. They are beneficial when pattern completion is crucial, such as restoring corrupted data or generating missing information.

III. RESULT AND DISCUSSION

The PowerShot G5 camera captured every image in the ALL-IDB dataset (Tab .1). Each image is a JPG file with a native resolution of 2592×1944 and a 24-bit colour depth. The images reflect various microscope magnifications (from 300 to 500). The ALL-IDB1 and ALL-IDB2 [18][19] versions of the database focus on the segmentation and categorization of lymphocyte cells, respectively.

TABLE I. DATASET DESCRIPTION

Image Acquisition Setup	
Camera	Canon PowerShot G5
Image format	JPG
Colour Depth	24 bit
Magnification of the Microscope	300 to 500
ALL-IDB1 ALL-IDB2	
Images	109260
Elements	39000260
Candidate Lymphoblasts	510130
Resolution	2592×1944×257×257

In the context of image analysis, including tasks like leukaemia classification, pre-processing involves a series of operations to enhance the quality of images, extract relevant features, and facilitate accurate and efficient analysis. The pre-processing of different approaches, namely linear, median, gaussian, salt and pepper, is illustrated in Fig. 1.

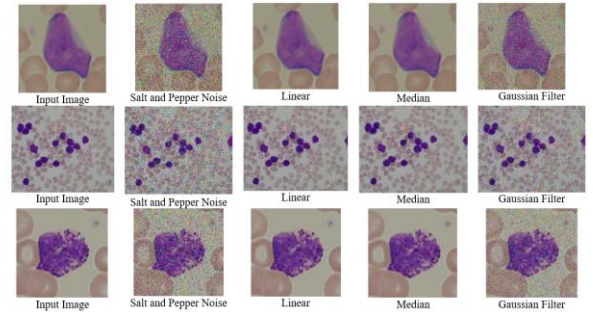


Fig. 1. Comparison of Preprocessing

Image analysis requires segmenting leukemia nuclei to identify and quantify cell nuclei. The segmentation of the nucleus using CNN, k-means clustering and FCM is illustrated in Fig. 2, and the comparative analysis is given in Fig. 3, where the FCM is accurate and quick compared to existing techniques. Segmentation time comparison among existing methods like CNN, k-means clustering, and proposed FCM can vary significantly based on factors such as image size, algorithm complexity, hardware, and implementation details. Research shows FCM is quick and efficient, with segmentation time displayed in Tab. 2.

Fig. 2 (a) inputs the leukaemia image, Fig. 2 (b) indicates the CNN, Fig. 2 (c) shows the k-means clustering, and Fig. 2 (d) is fuzzy-c-means clustering.

FCM appears to be the most efficient in terms of segmentation time [22]. K-means clustering follows FCM, while CNN exhibits the highest segmentation times, making it the least efficient method [23]. CNN tends to have the highest segmentation time among the three ways. It is expected since CNNs are computationally more intensive due to their architecture and the training required [20]. Both k-means clustering and FCM show increasing segmentation times as the image count grows. This is likely because the number of data points and clusters generally influences the computational complexity of these methods.

TABLE II. COMPARISON OF SEGMENTATION TIME

Image Count	CNN	k-means clustering	FCM
50	5765	7852	4523
100	6871	8256	4927
150	7891	8927	5239
200	8267	9762	5927

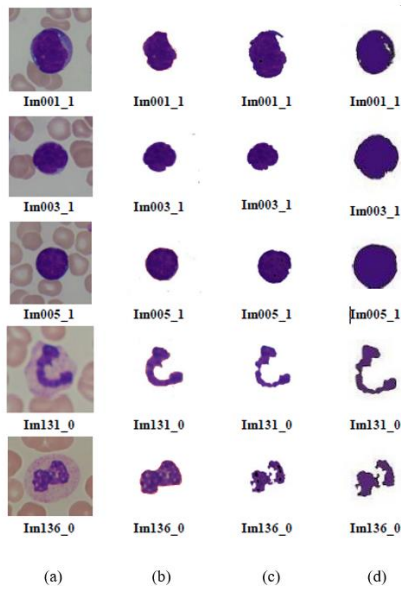


Fig. 2. Comparison of Segmentation

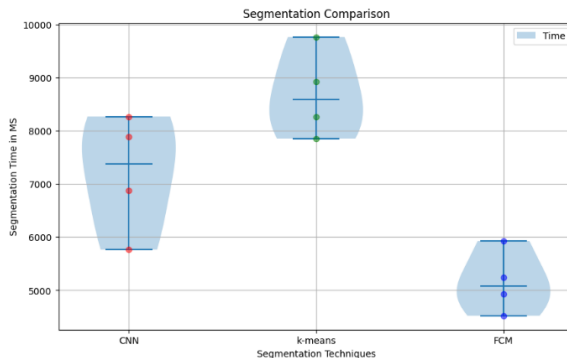


Fig. 3. Comparison of Segmentation Time

Classification accuracy in leukemia typically refers to how well a DL model can correctly classify leukemia cases based on input features or images. By comparing predicted instances to the total dataset, classification accuracy ensures accurate predictions and high accuracy (Tab. 3)

$$\text{Accuracy} = \frac{\text{Count of Correct Prediction}}{\text{Total Prediction}} \quad (5)$$

From Fig. 4, BAMNN is a promising method with consistently high accuracy [21]. Further research and experimentation could explore optimising and fine-tuning the BAMNN architecture to achieve even better results.

TABLE III. COMPARISON OF ACCURACY

Image Count	LDSVM	CNN	BAMNN
50	85.89	89.67	92.67
100	86.56	90.54	95.45
150	87	91.67	95.78
200	88.91	92	96.87

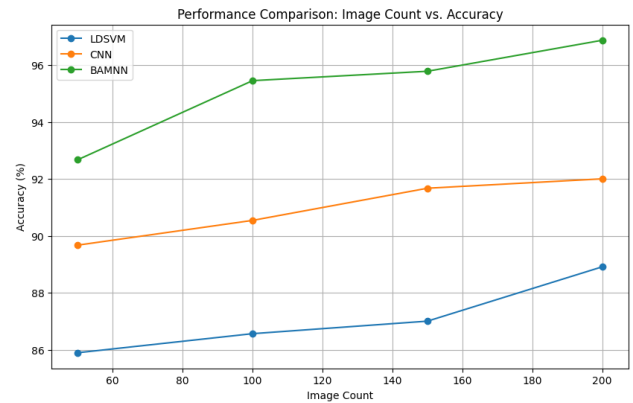


Fig. 4. Comparison of Accuracy

IV. CONCLUSION AND FUTURE WORK

In this research endeavour, the research explored the application of Bidirectional Associative Memory Neural Networks (BAMNN) for image classification. This novel approach combines the associative memory capabilities of BAMs with the power of neural networks. The investigation delved into adapting BAMNNs to the image classification task, demonstrating its potential to address classification challenges while leveraging its bidirectional associative properties. The conceptual model and preliminary experimentation showed promising outcomes in capturing image-to-class associations using BAMNNs. Integrating associative learning principles into neural network architectures opens new chances for advancing image classification methodologies. The research findings underline the potential of BAMNNs to provide robust classification results while enhancing the interpretability of learned associations. Several exciting directions emerge for further exploring and enhancing the BAMNN approach in image classification.

In future, BAMNNs can be extended to multi-class classification scenarios by adapting the architecture and learning rules to accommodate multiple class labels. The researcher can develop approaches to handle class imbalances and explore ensembling techniques for improved performance.

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